



ESB Technical Interview: Modular Prosthetic Gripper

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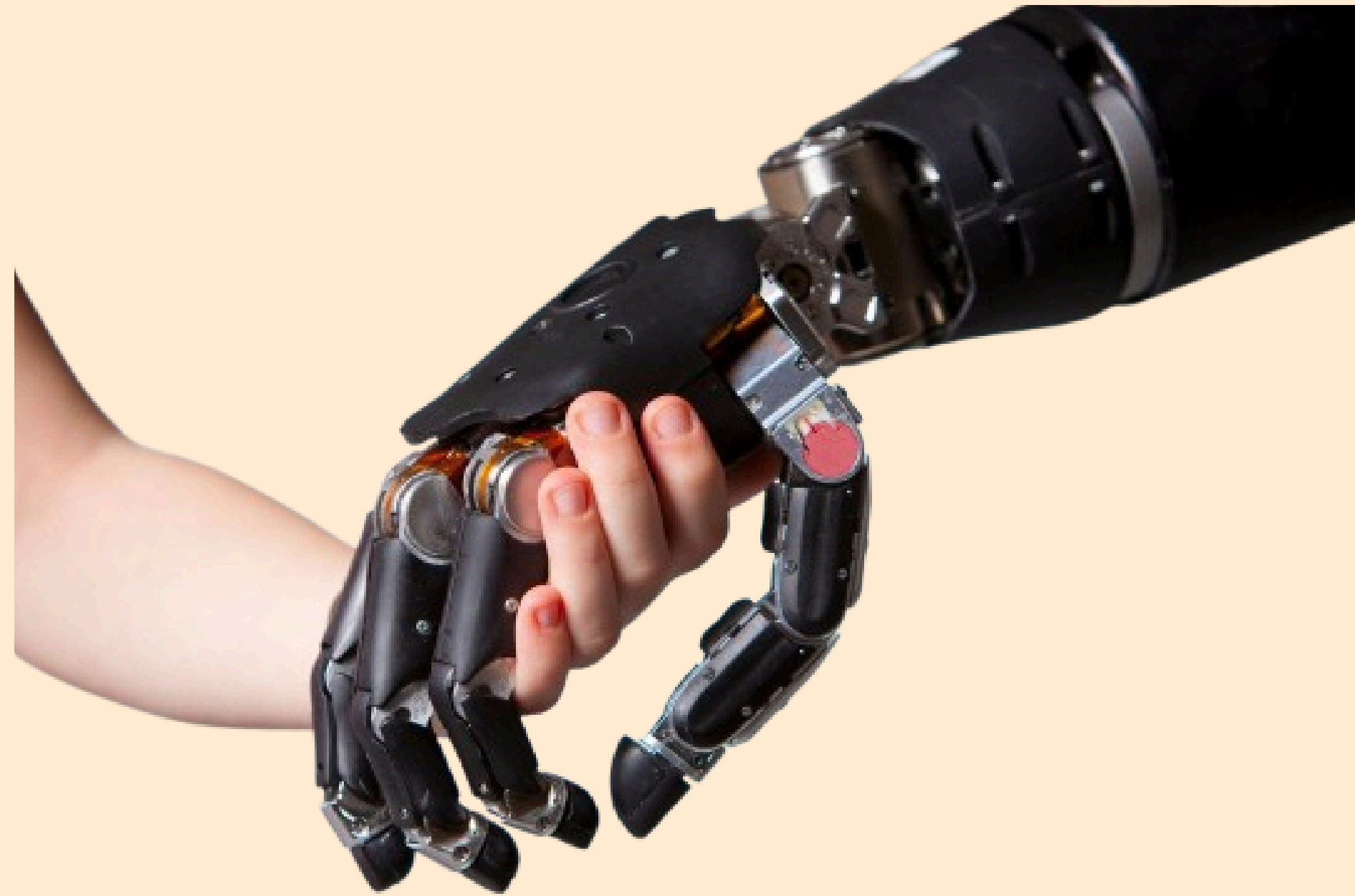


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Prompt: Design a **modular prosthetic gripper** to attach to a prosthetic limb for **improving precision and feedback** for rehabilitation (provides fine motor control but also integrates sensor feedback)





Assumptions:

1. Myoelectric prosthetic limb
 2. Battery in prosthetic limb → prosthetic gripper
 3. Regular limb design (except for modularity)
 4. Electric modular gripper
- 

Factors to Consider

COST

MANUFACTURABILITY

EASE OF USE/COMFORT (REHAB)

MODULARITY

SENSOR FEEDBACK

Topics Addressed



Hand Model/Material

- Material of the “bone”
- Design of the Hand

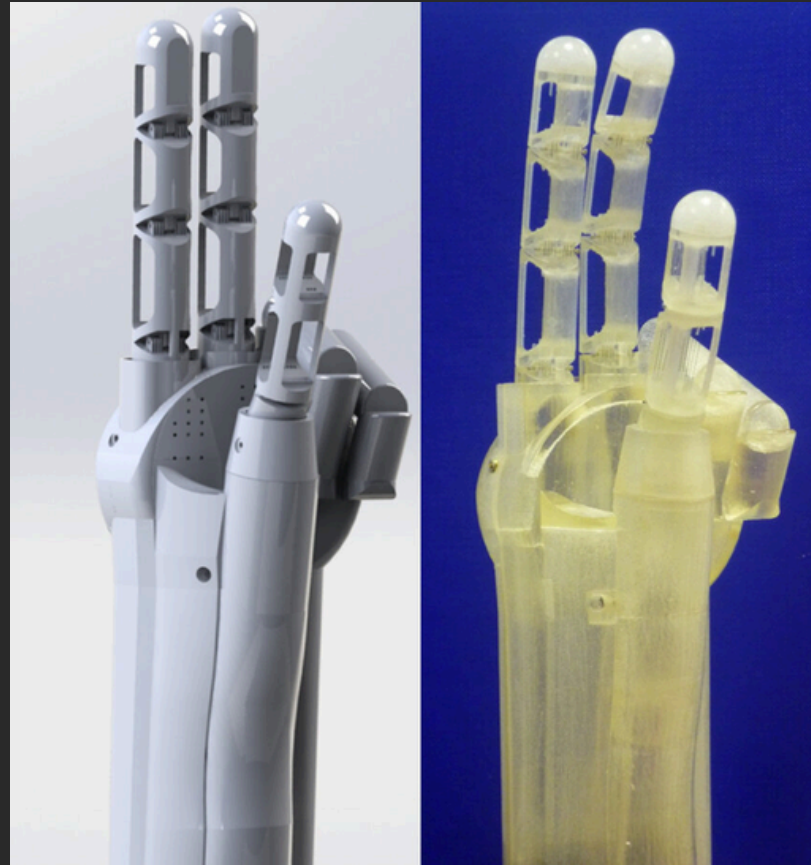
Internal Components

- Sensors
- Motors
- Microcontroller
- Feedback from sensors

Modularity.

- Physical connection to limb
- Access to power
- Data from myoelectric sensors

3 Finger Model



5 Finger Model



Clamp Model



Category	Clamp Hand	3-Finger Gripper	5-Finger Gripper
Dexterity	1	2	3
Strength & Durability	1	3	2
Weight (lightest = 3)	3	2	1
Complexity (higher = more complex)	1	2	3
Cost (affordable = higher rank)	3	2	1
Ease of Training	3 (simple, educational)	2	1 (steep learning curve)
Realistic Appearance	1	2	3
Best for Rehabilitation Stage	1 (very basic)	3 (early rehab, daily use)	2 (advanced rehab, complex use)

Choice: 3 Finger Model

Strongest balance of durability, cost, weight, and function



Material for Prosthetic Gripper

- **Strength, weight, durability, and biocompatibility**

Material for Prosthetic Gripper

TITANIUM

Pros: Lightweight, excellent strength-to-weight ratio, strong corrosion resistance, biocompatibility

Cons: Expensive, difficult to machine

Material for Prosthetic Gripper

ALUMINUM

Pros: Lightest, good corrosion resistance,
low cost, easy machinability

Cons: Moderate strength

Material for Prosthetic Gripper

STAINLESS STEEL

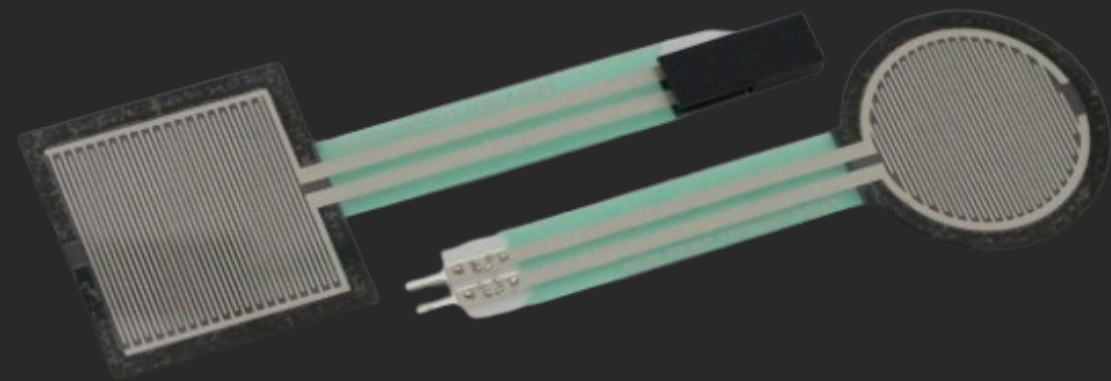
Pros: High strength/durability, good corrosion resistance

Cons: Heaviest, moderate cost, fair machinability

Choice: Aluminum



Sensors Used



TACTILE SENSORS

- Piezoelectric and Piezoresistive sensors
- Anti-slip/grip adjustments

FORCE/PRESSURE SENSORS

- Force Sensing Resistors (FSRs)
- Micro-Measurements Strain Gauge Sensors
- Detect applied force
- Auto-grip adjustments

Attribute	BLDC (Brushless DC)	Coreless DC	Micro Gear DC	Ultrasonic Motor
Torque	High torque density; good across wide speeds	Moderate for size; very smooth	Very high <i>output</i> torque (due to gears), low speed	High holding/low-speed torque, limited at continuous high speed

Attribute	BLDC (Brushless DC)	Coreless DC	Micro Gear DC	Ultrasonic Motor
Response time	Fast (good acceleration)	Very fast (low inertia)	Slower; gearbox adds lag	Extremely fast, near-instant response

Attribute	BLDC (Brushless DC)	Coreless DC	Micro Gear DC	Ultrasonic Motor
Precision	Good with encoder/FOC; moderate open-loop	Good analog proportionality; best with encoder	Limited by gear backlash; needs encoder	Excellent, can reach sub-micron; self-locking

Attribute	BLDC (Brushless DC)	Coreless DC	Micro Gear DC	Ultrasonic Motor
Size	Compact for given power, scalable	Very compact, especially small diameters	Very small motor, gearbox adds bulk	Thin “pancake” or slim forms

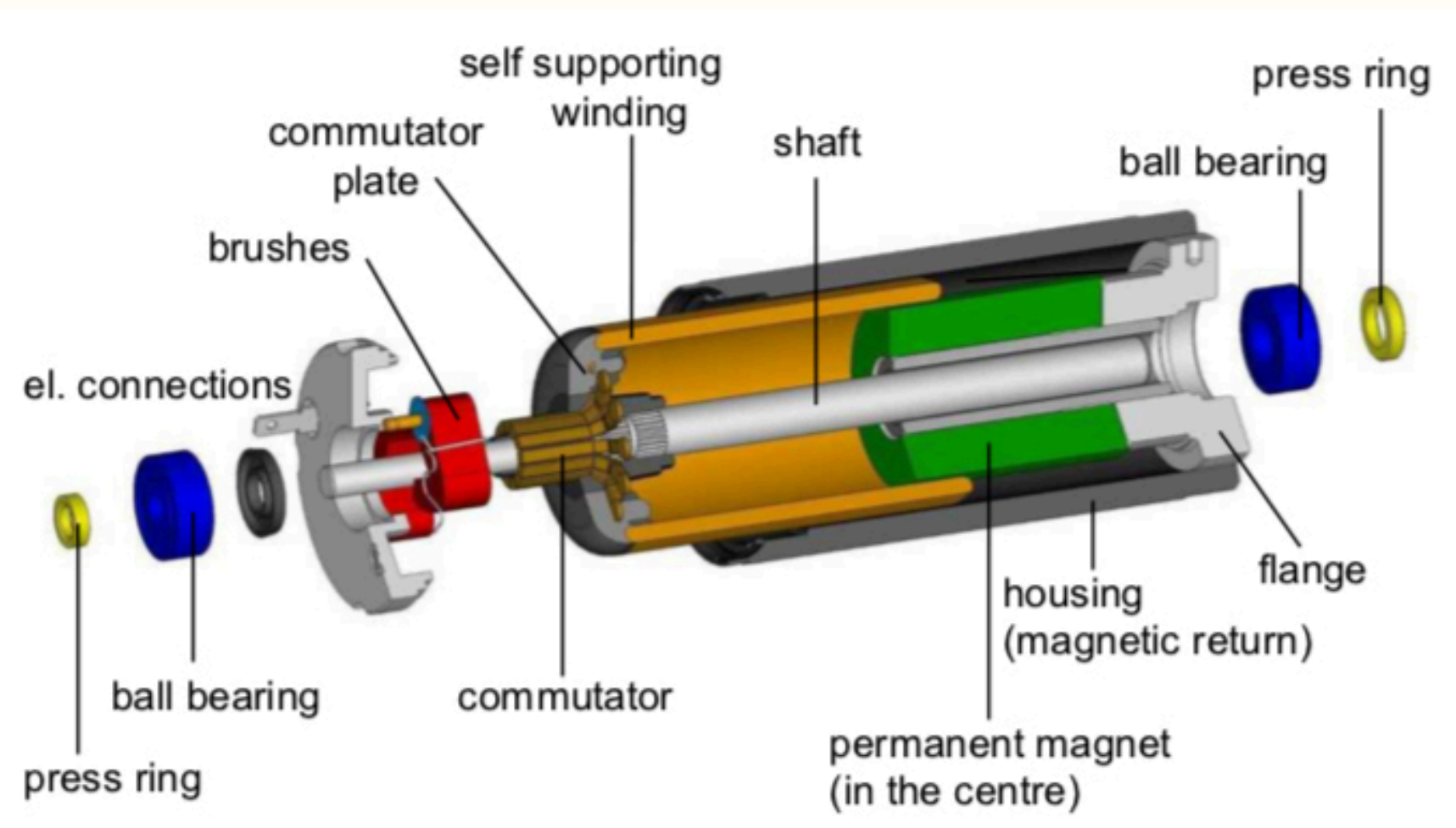
Attribute	BLDC (Brushless DC)	Coreless DC	Micro Gear DC	Ultrasonic Motor
Cost	\$\$-\$\$\$ (requires driver)	\$\$ (simple driver)	\$-\$\$ (common, cheap)	\$\$\$ (specialized)

Attribute	BLDC (Brushless DC)	Coreless DC	Micro Gear DC	Ultrasonic Motor
Energy efficiency	High (80–90%)	Medium–high	Low–medium (losses in brushes + gears)	Low–medium continuous, efficient for holding

Attribute	BLDC (Brushless DC)	Coreless DC	Micro Gear DC	Ultrasonic Motor
Weight	Medium	Very light	Light–medium (gearbox adds)	Light and thin

Choice: Coreless DC Motors

Compact & lightweight
High torque
Fast response .
Low cost
Easy Control





Micro-controller

Doesn't need microprocessor

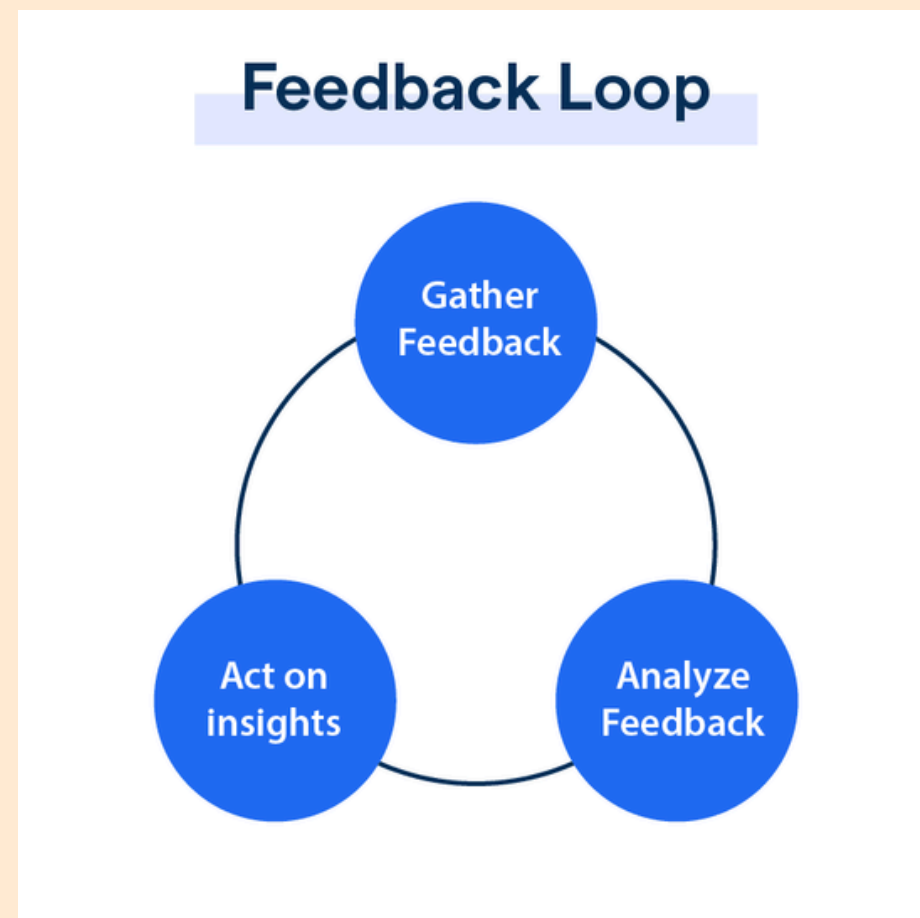
Low-level tasks

Recommended: STM32



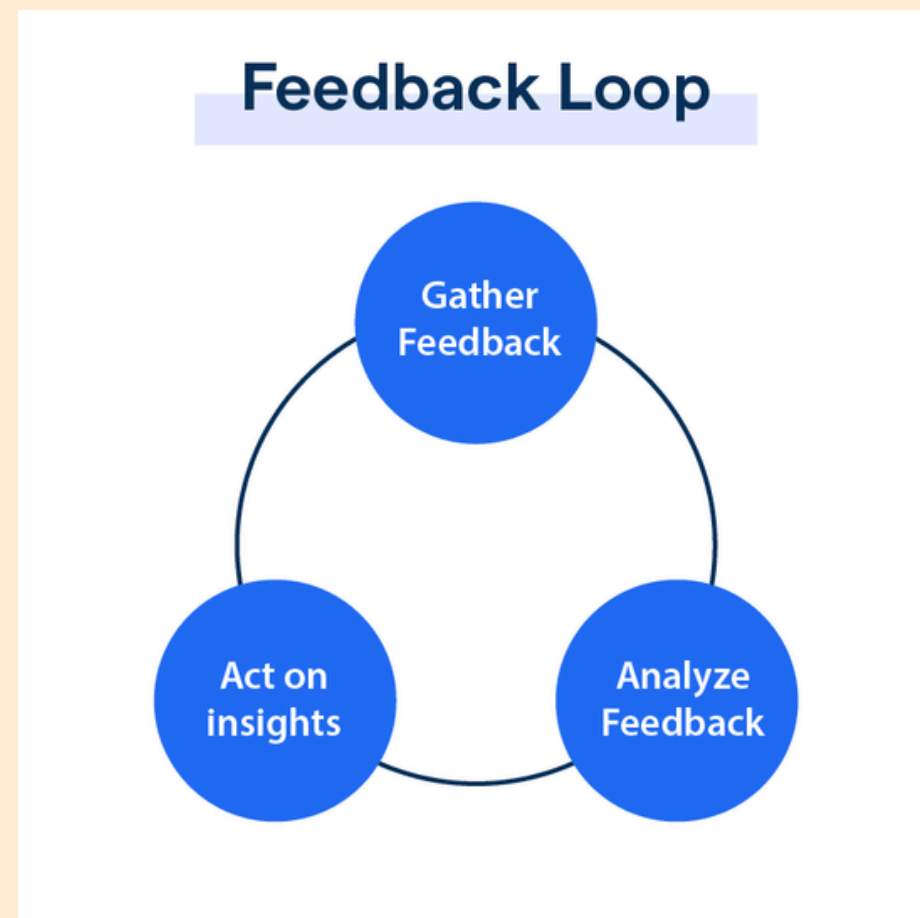
Feedback Loop

Piezoelectric and piezoresistive detect slipping → notify microcontroller to tighten grip slightly/increase force limit (originally light system)



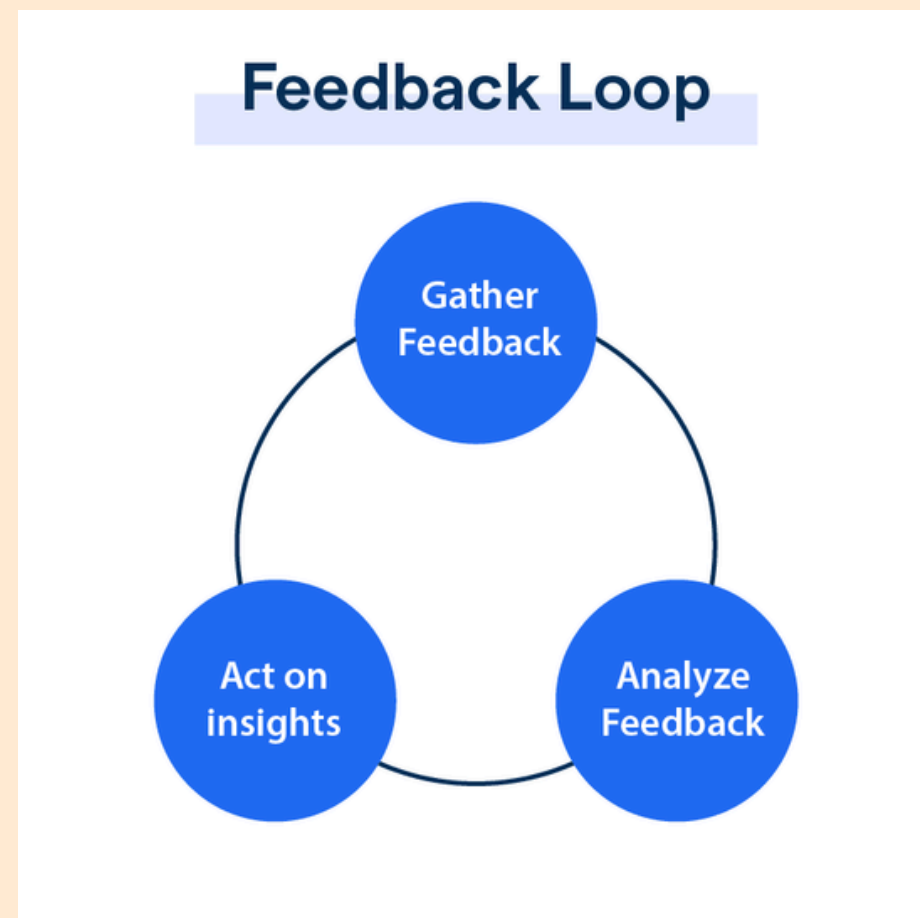
Feedback Loop

Piezoresistive resistors detects hardness/softness of object → change grip speed/force



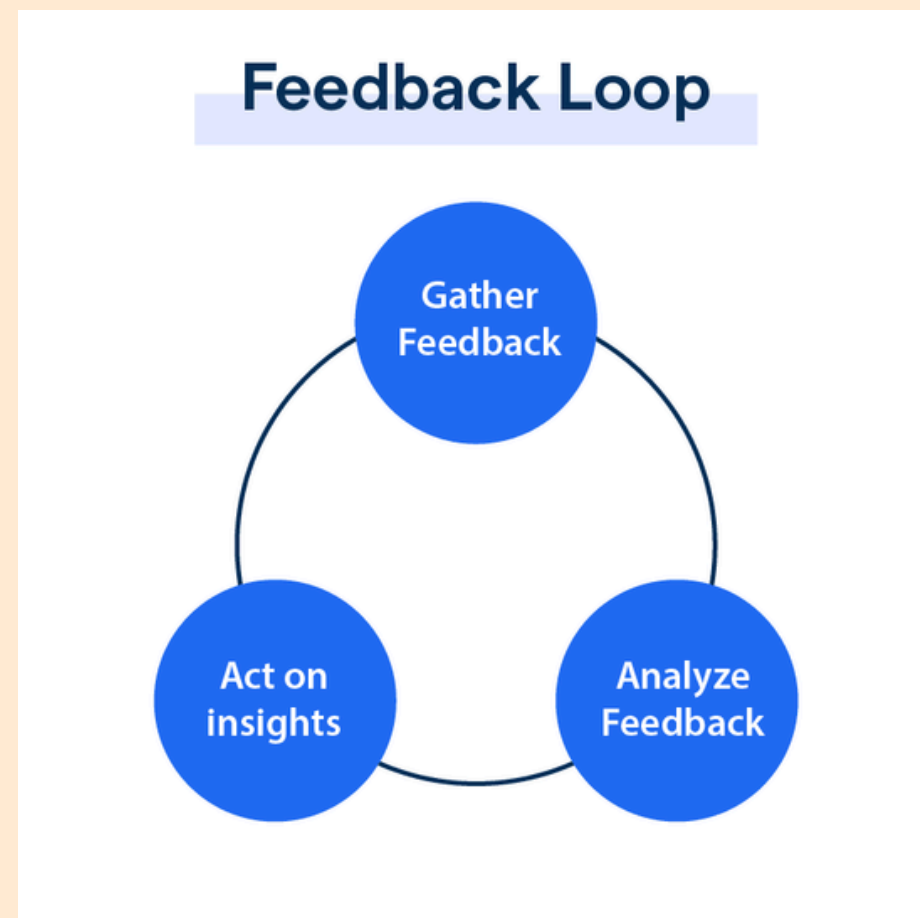
Feedback Loop

FSRs detect contact to prevent fingers from closing too quickly (sets initial limit for force)



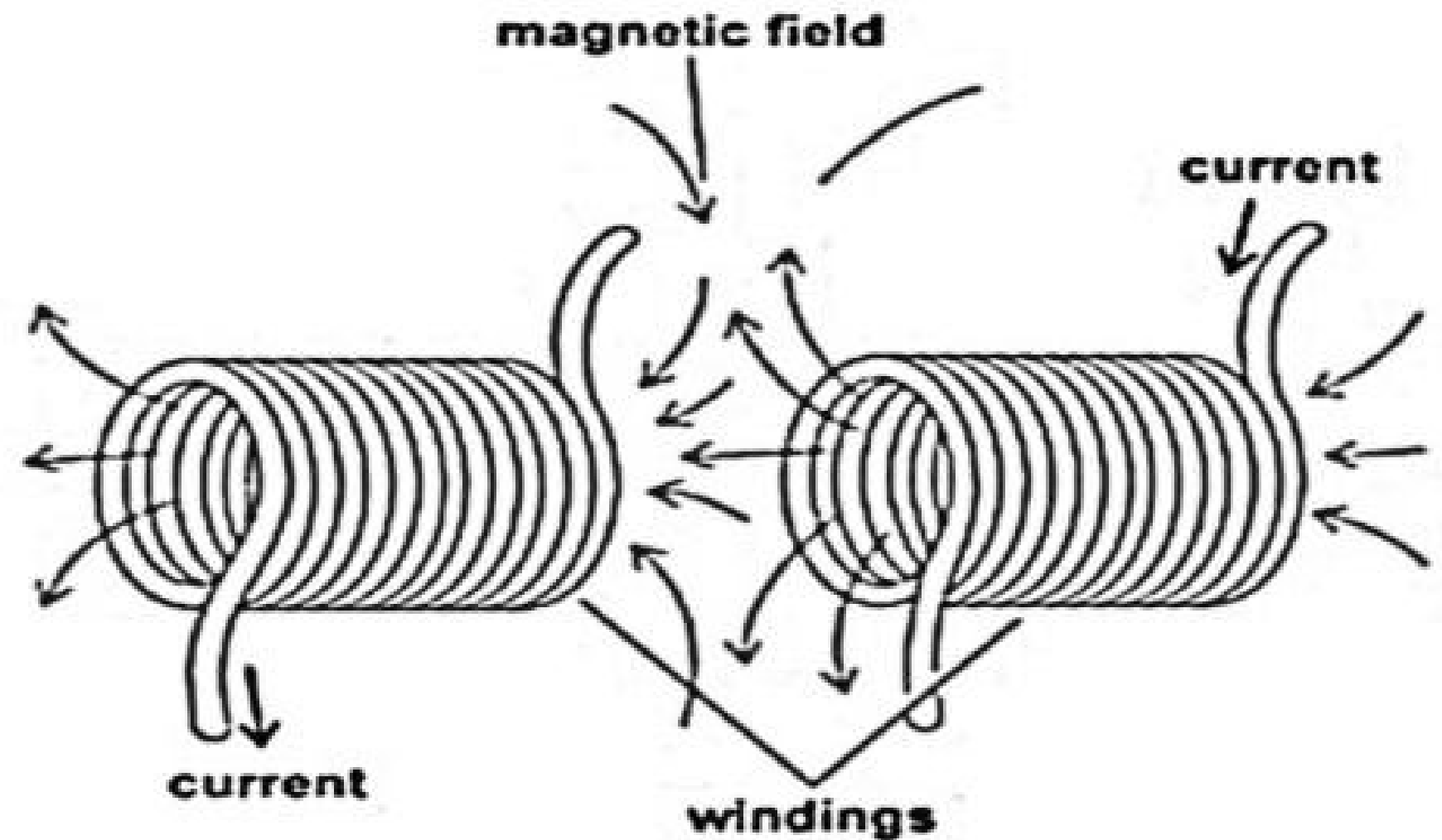
Feedback Loop

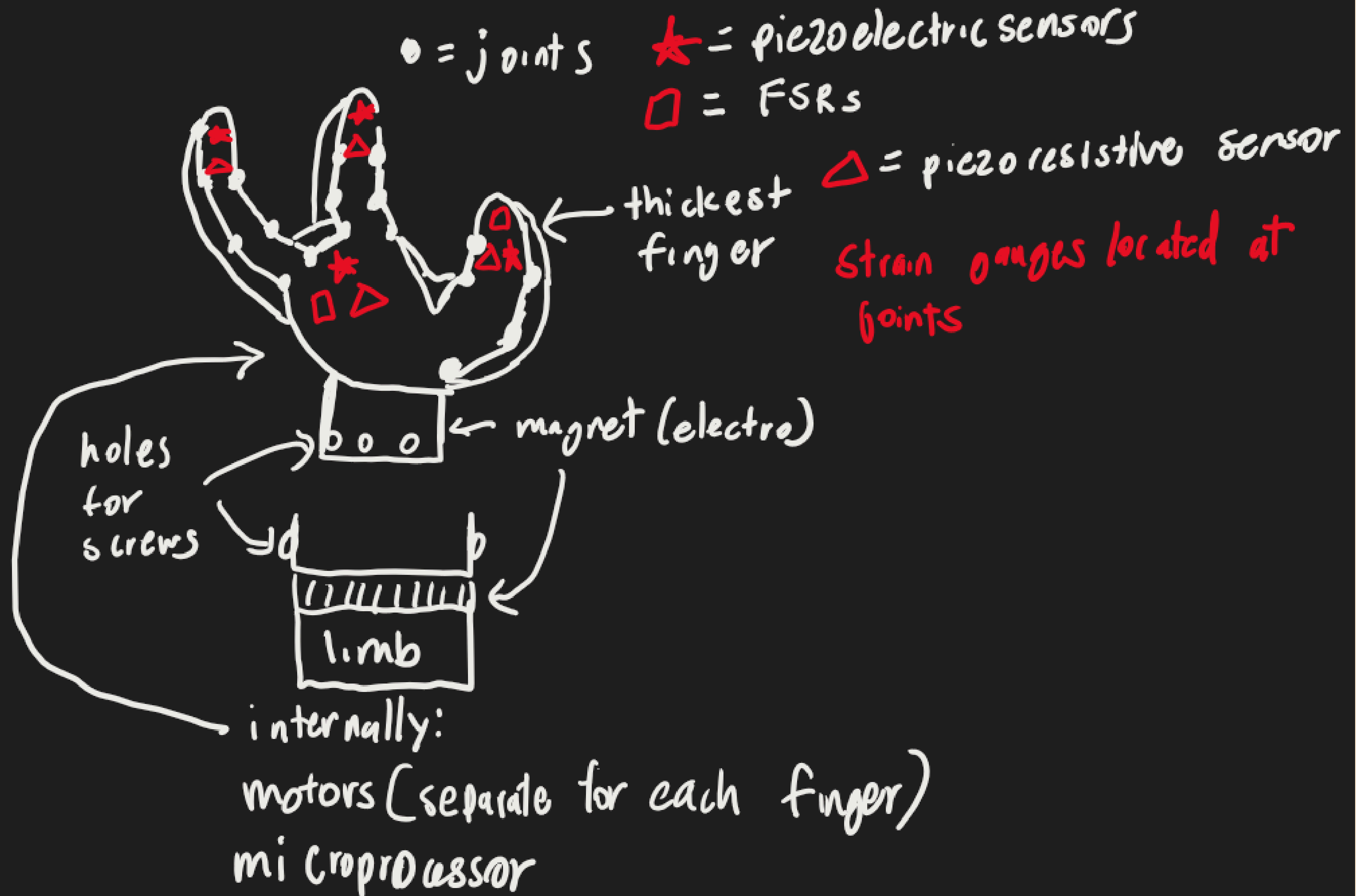
Eccentric Rotating Mass Motors for vibration feedback (double when slipping, triple when at force limit)



Modularity

- Electromagnetic connection
- Screw holes
- Inductive or magnetic coupling





Main Problems Faced

FORCE LIMITATIONS

How to prevent objects from being gripped too hard or too lightly? (e.g. glass)

MODULARITY

How to create something that can connect with the limb and its myoelectric sensors?

FEEDBACK LOOP

How to deliver feedback and adapt to it?



Future Plans



- Modular Fingers
- AI (grip recognition)
- Five-finger model
- Micro-processor
- Macros

THANKS!

